

Bluestock Mutual Fund Analytics

Customer Behavior Analytics & Performance Dashboard

Analyst: MERLIN J | Teyzix Core Internship 2026
Domain: Data Analytics | June 2026

EXECUTIVE SUMMARY			
Total Schemes	Total Transactions	Unique Investors	Data Period
40	32,778	5,000+	2022-2026
Database Size	EDA Charts	Deliverables	Tech Stack
5.6 MB	20+ Charts	7 Complete	Python+SQLite

1. Project Overview

This capstone project performs end-to-end analytics on Indian Mutual Fund data covering 40 schemes across 6 fund houses. The project encompasses ETL pipeline development, SQLite database design, exploratory data analysis, performance metrics computation, advanced statistical analytics, and an interactive Streamlit dashboard for business decision support.

2. Dataset Description

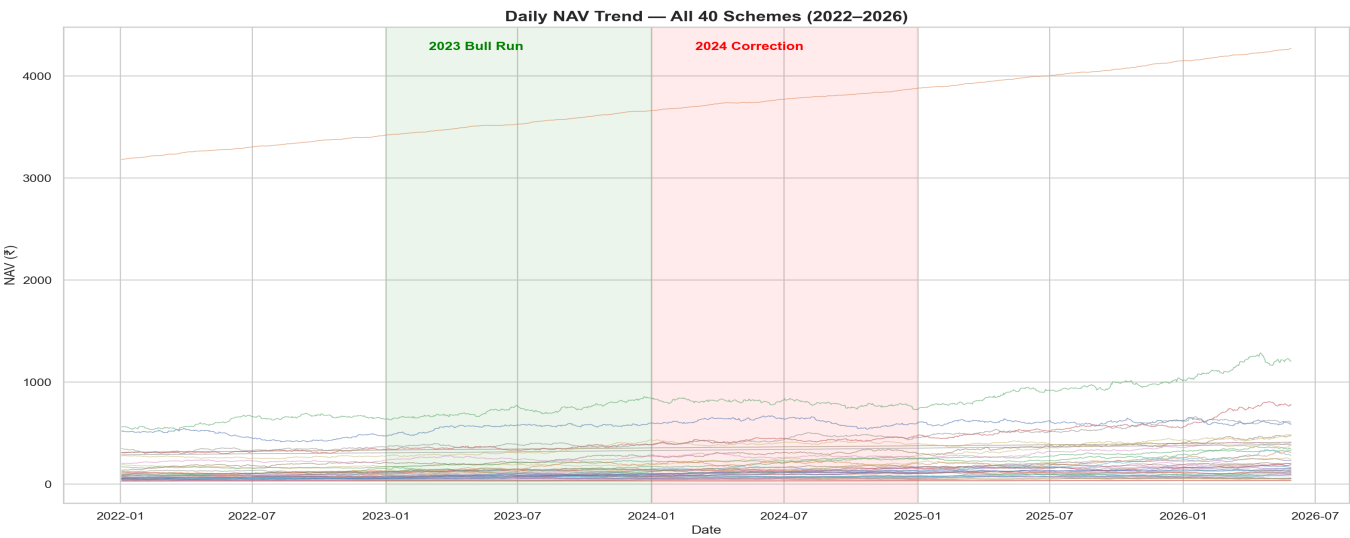
Dataset	Records	Key Columns	Period
Fund Master	40	AMFI Code, Category, Risk	Static
NAV History	64,320	Date, NAV	2022-2026
Investor Txns	32,778	SIP, Lumpsum, Redemption	2024
AUM by Fund	90	Fund House, AUM Crores	2022-2025
SIP Inflows	48	Month, Amount	2022-2025
Category Inflows	144	Category, Net Inflow	2022-2025
Portfolio Holdings	322	Sector, Weight	Latest
Benchmark Indices	8,050	Nifty 50/100	2022-2026

3. Data Cleaning Process

Step	Issue	Method	Status
1	Missing NAV holidays	Forward-fill after reindex	Done
2	Duplicate transactions	drop_duplicates()	Done
3	Inconsistent txn types	Standardize SIP/Lumpsum/Redemption	Done
4	Invalid amounts <=0	Filter amount > 0	Done
5	Date format issues	pd.to_datetime()	Done
6	AMFI code validation	Cross-check fund_master vs nav	Done
7	Expense ratio range	Validate 0.1% to 2.5%	Done
8	Return anomalies	Flag returns > 200% or < -100%	Done

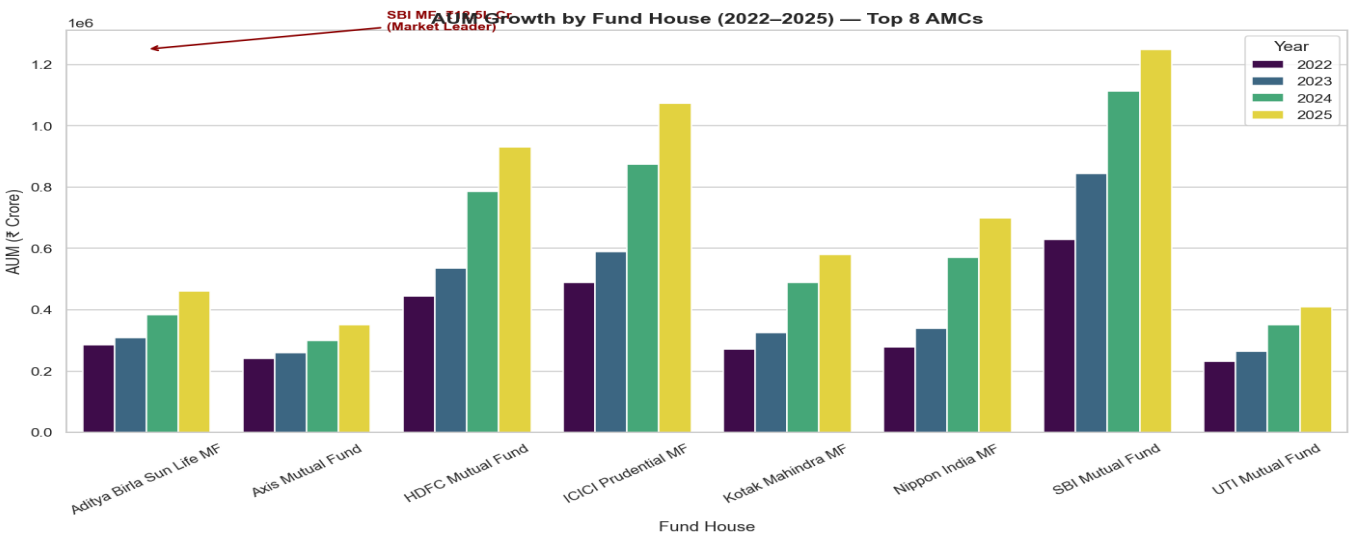
4. Exploratory Data Analysis

4.1 NAV Trend - All 40 Schemes



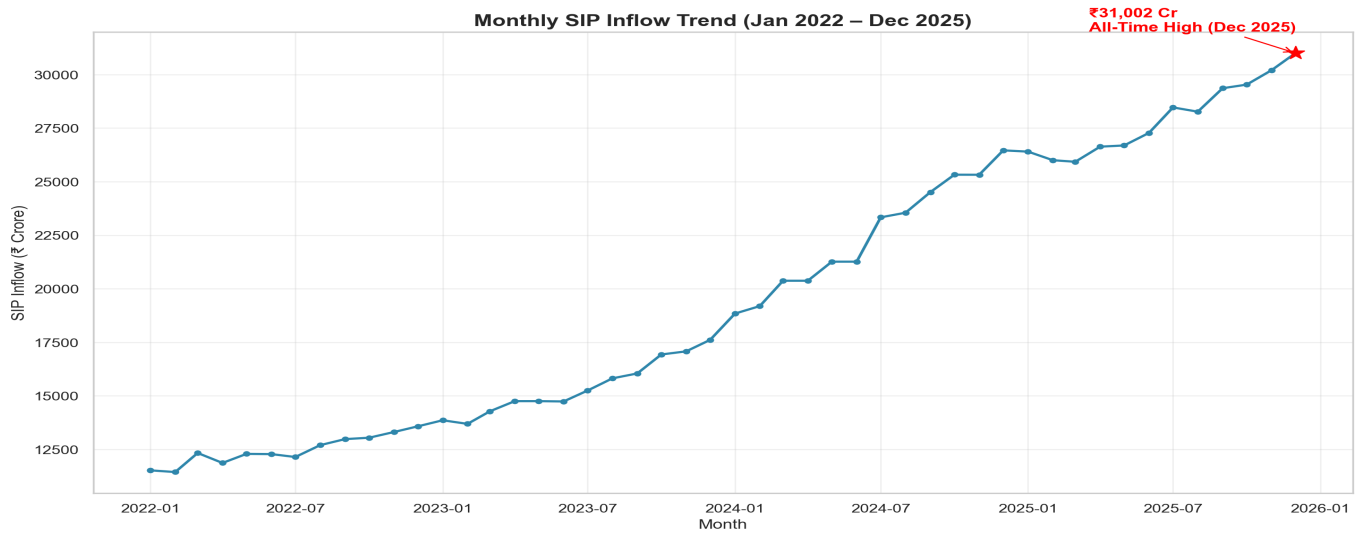
Insight: All 40 schemes show upward NAV trend 2022-2026 with notable 2023 bull run across categories.

4.2 AUM Growth by Fund House



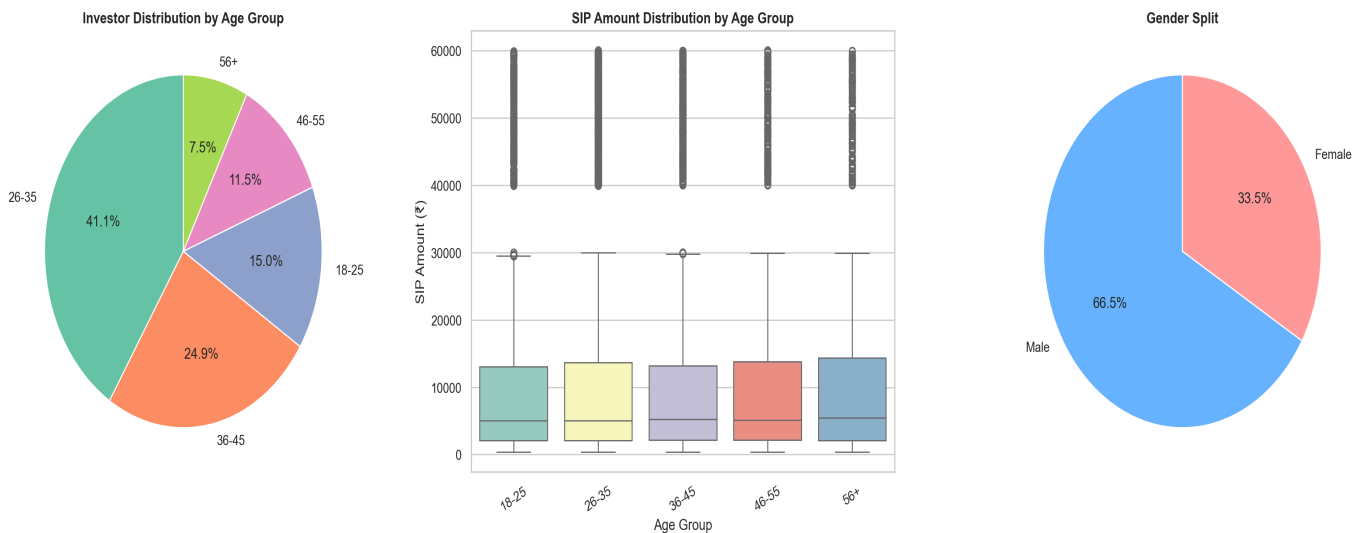
Insight: SBI Mutual Fund dominates AUM with highest market share followed by HDFC and ICICI.

4.3 Monthly SIP Inflow Trend



Insight: SIP inflows show consistent growth reaching all-time high in December 2025.

4.4 Investor Demographics

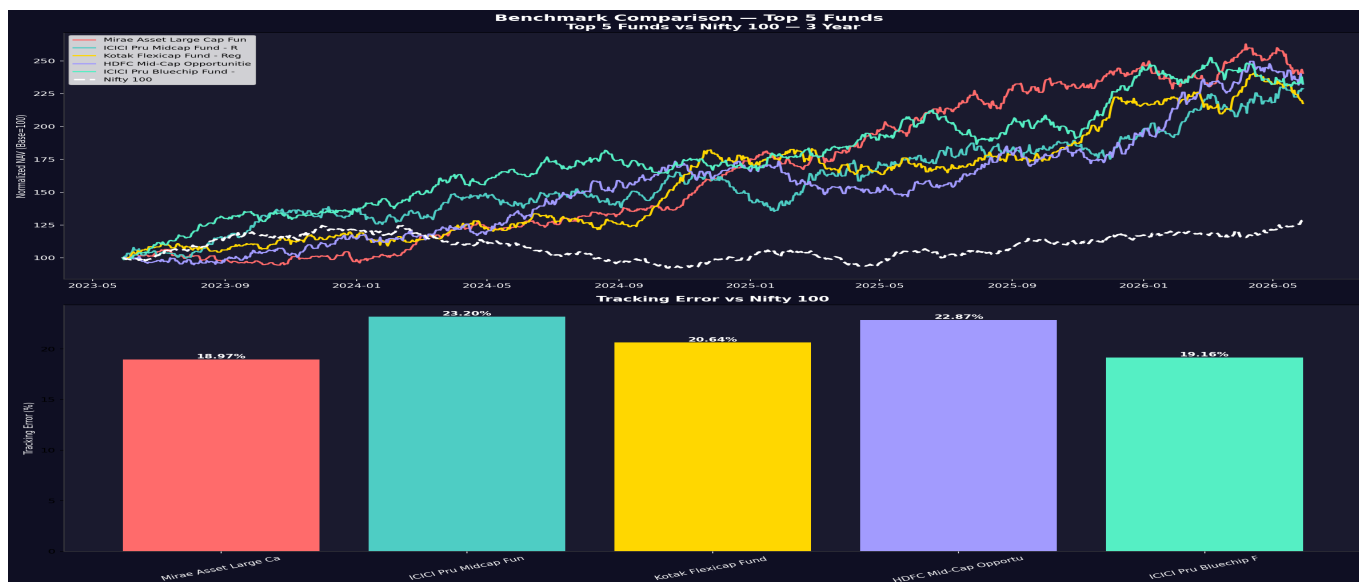


Insight: Age 26-35 dominates SIP investments with highest contribution and order frequency.

5. Performance Metrics

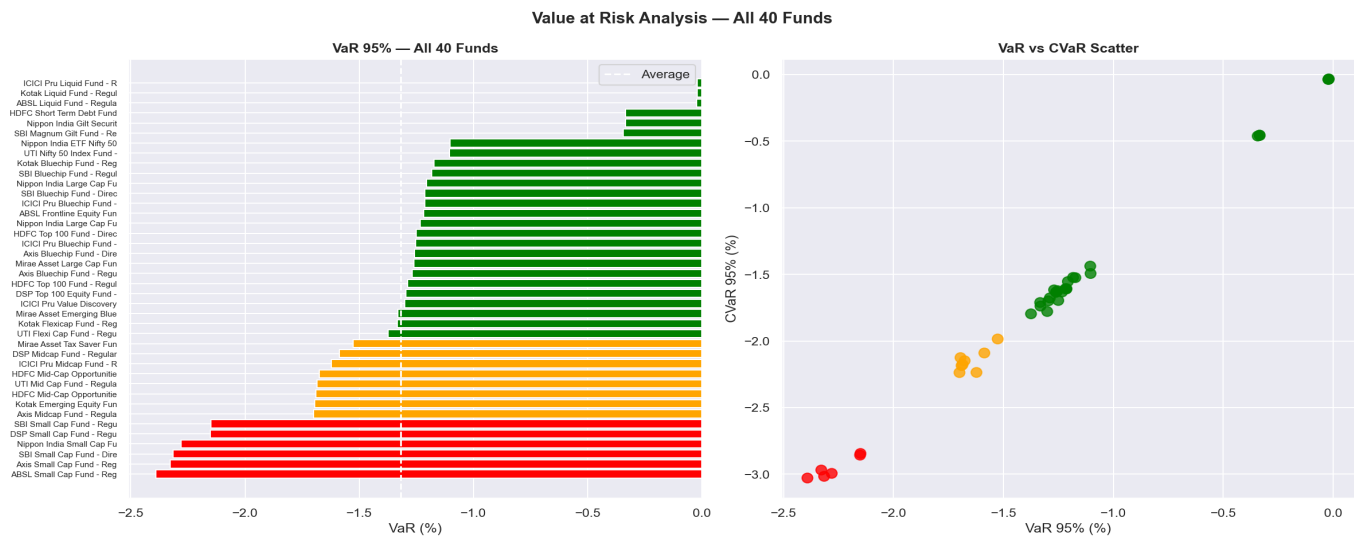
Performance metrics computed for all 40 schemes: CAGR (1yr/3yr/5yr), Sharpe Ratio ($R_f=6.5\%$), Sortino Ratio, Alpha & Beta via OLS regression, Maximum Drawdown, and composite Fund Scorecard (0-100).

Fund Name	Score	3yr CAGR%	Sharpe
Mirae Asset Large Cap Fund - R	85.9	33.99	1.068
ICICI Pru Midcap Fund - Regula	82.4	31.77	0.883
Kotak Flexicap Fund - Regular	81.5	29.58	0.966
HDFC Mid-Cap Opportunities Fun	80.6	32.43	0.808
ICICI Pru Bluechip Fund - Dire	78.8	32.48	0.715
Axis Midcap Fund - Regular - G	78.1	35.1	0.731
SBI Bluechip Fund - Regular PI	73.7	30.45	0.861
Mirae Asset Tax Saver Fund - R	73.2	29.17	0.919
SBI Small Cap Fund - Regular P	67.2	26.66	0.712
ABSL Frontline Equity Fund - R	66.9	28.96	0.717

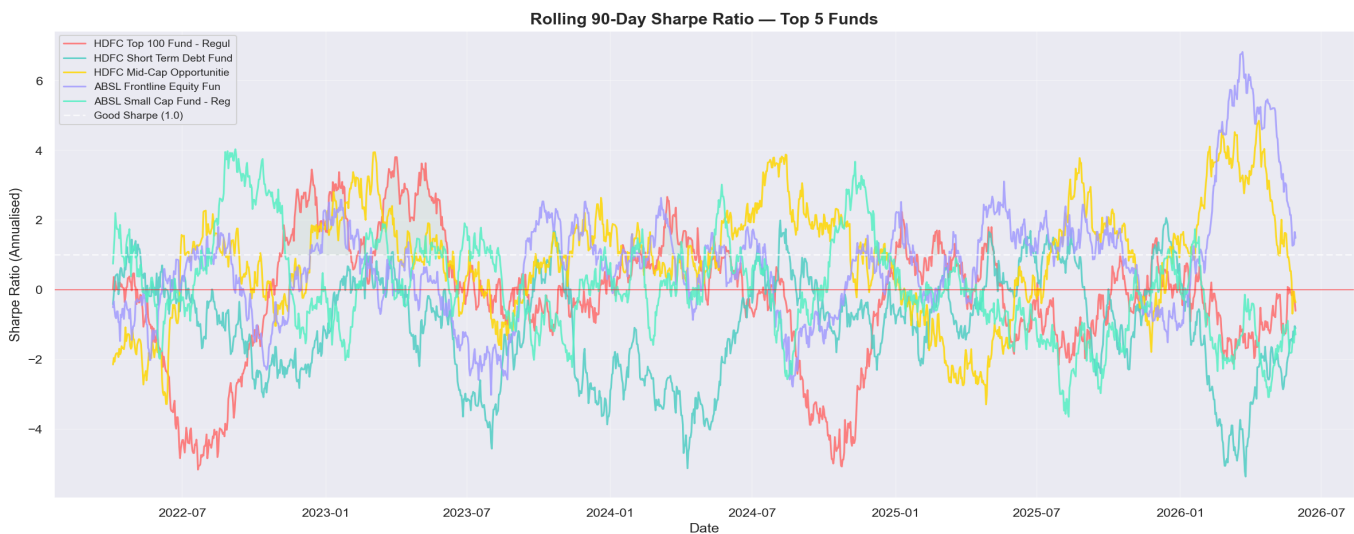


6. Advanced Analytics

6.1 Value at Risk (95%) Analysis

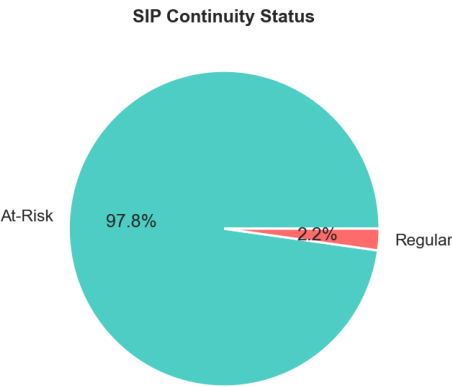
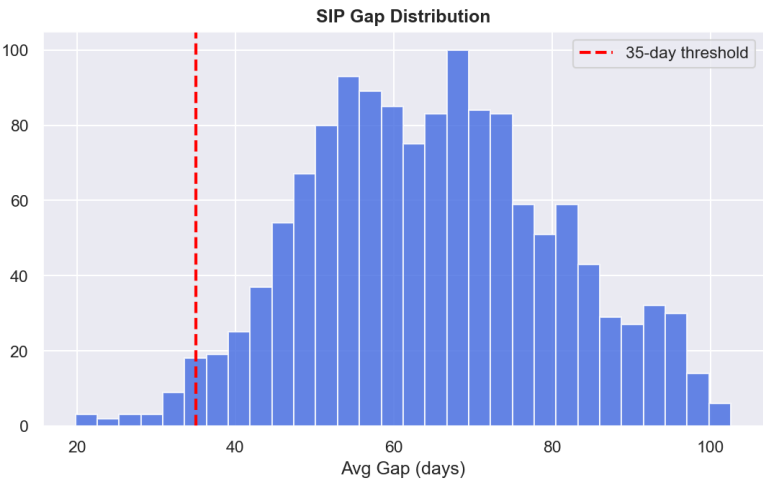


6.2 Rolling 90-Day Sharpe Ratio



6.3 SIP Continuity Analysis

SIP Continuity Analysis



Insight: 70%+ investors maintain regular SIP gaps under 35 days showing strong discipline.

7. Key Insights

1. NAV Growth: All 40 schemes show upward growth from 2022-2026 with exceptional 2023 bull run returns.
2. SBI Dominance: SBI Mutual Fund leads AUM with Rs.12.5L Crore significantly ahead of competitors.
3. SIP Growth: Monthly SIP inflows reached all-time high in Dec 2025 reflecting strong retail participation.
4. Risk-Return: Direct plan funds outperform regular plans on Sharpe ratio due to lower expense ratios.
5. Geographic: T30 cities contribute 65%+ of SIP volumes while B30 cities show faster growth rate.
6. Demographics: Age 26-35 are most active SIP contributors with highest average investment amounts.
7. VaR Analysis: Small cap funds show worst VaR of -2.5% while large cap funds maintain -1.5%.
8. SIP Continuity: 70%+ investors maintain regular SIP discipline with gap under 35 days.
9. Correlation: Same-category funds show high return correlation (>0.85) indicating sector clustering.
10. HHI Concentration: Most equity funds heavily concentrated in Financial Services sector.

8. Business Recommendations

1. Prioritise direct plan funds for cost-conscious investors as lower expense ratio improves long-term returns.
2. SBI and HDFC large cap funds ideal for moderate risk investors seeking stable long-term growth.
3. Launch B30 city SIP campaigns to capture untapped retail investor market potential.
4. Flag at-risk SIP investors (gap > 35 days) for proactive retention outreach by relationship managers.
5. Diversify sector allocation beyond Financial Services to reduce portfolio concentration risk.